

RAI-853: Advanced Robotics with ROS

Lec 01: Introduction

Instructor:

Dr. Khawaja Fahad Iqbal

Deputy HoD (Department of Robotics & AI, SMME, NUST)

Scientific Director (National Center of Artificial Intelligence)



- **My Introduction**
- **Course Introduction**
- **Your Introduction**

- **My Introduction**
- Course Introduction
- Your Introduction

Dr. Khawaja Fahad Iqbal

Deputy Head of Department

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- **Experience:**

- Deputy HoD (Sep 2025 - Present)
Department of Robotics and Artificial Intelligence, NUST.
- Scientific Director (Jul 2022 - Present)
Intelligent Robotics Lab (IRL), National Center of Artificial Intelligence (NCAI).
- CEO Rise AI Pvt. Ltd. (Jan 2023 - Present)
National Science and Technology Park (NSTP), NUST.
- Assistant Professor (Sep 2022-Present)
Department of Robotics and Artificial Intelligence, NUST.

- **Research:**

- 122M+ PKR (430k+ USD) Funding
- 30+ Research Publications
- Research lab with 30+ students



Dr. Khawaja Fahad Iqbal

- **Education:**

- PhD Robotics (2022), Tohoku University, Japan.
- MS Bioengineering and Robotics (2017) , Tohoku University, Japan.
- BE Mechatronics Engineering (2011), NUST, Pakistan.

- **Extra-Curricular Activities**

- Treasurer (2024-Present)
IEEE Robotics and Automation Society Pakistan
- Finance Secretary (2024-Present)
MEXT Alumni Association Pakistan
- Member National AI Task Force (2023 - Present)



Dr. Khawaja Fahad Iqbal

Institute: College of E&ME, NUST.

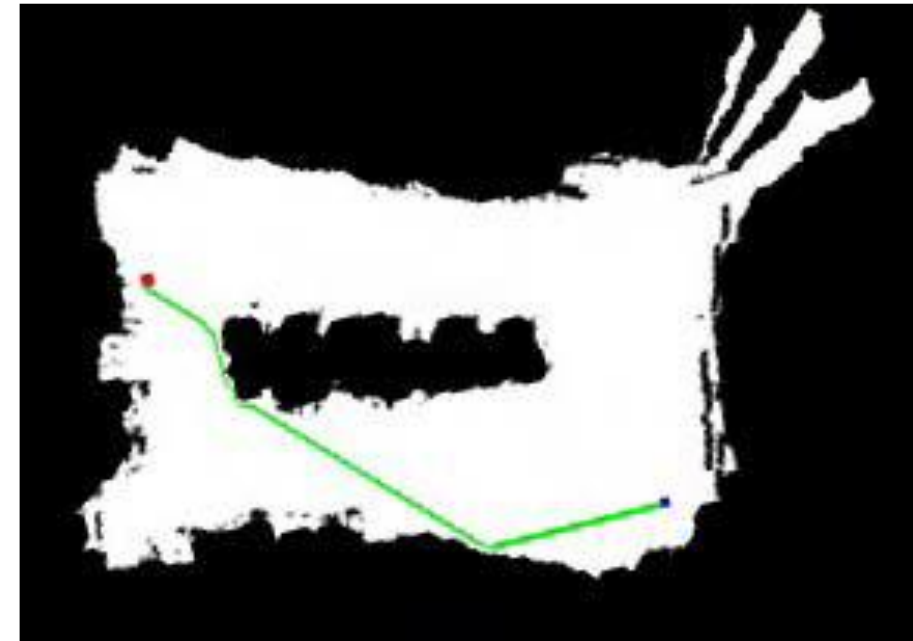
Scholarship: NUST High Achievers Scholarship.

Thesis Topic: Intelligent DRIVING System (IDRIS).

Funding: AFIRM, Rawalpindi, Pakistan.

Supervisors: Dr. Kunwar Faraz, Dr. Yasar Ayaz.

Key Areas: Mobile Robots, Motion Planning, Neural Networks.



Our algorithm finds the optimal path between the robot's current position and the goal position.

Institute:

RISE Lab and Research Center, Department of Robotics and Artificial Intelligence, SMME, NUST.

Labs Conducted:

1. Logic Design and Microprocessors (ME-02), Spring 2013, with Lec. Jawad Aslam.
2. Electrical Engineering (ME-03), Spring 2013, with Dr. Mohsin Jamil.
3. Computer Systems and Programming (ME-05), Fall 2013, with Dr. Omer Gilani.
4. Mobile Robots (RIME-12), Fall 2013, with Dr. Yasar Ayaz.
5. Engineering Design and Graphics (ChE-06, MtlE-06, from SCME), Spring 2014, with Lec. Adnan Hanif.
6. Electrical Engineering (ME-04), Spring 2014, with Dr. Umar Ansari.

Project Supervision:

1. Design and Development of a Robotic Mannequin.
2. Perceptual Aliasing Solutions for Simultaneous Localization and Mapping – SLAM.
3. Human Tracking and Following for Service Robots.
4. RRT* based Motion Planning for Mobile Robots.
- 5. Sakura Wheelchair Project.**
- 6. Robocup Standard Platform League – SPL.**

Institute: Tohoku University, Japan.

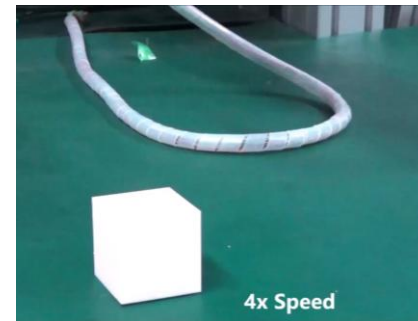
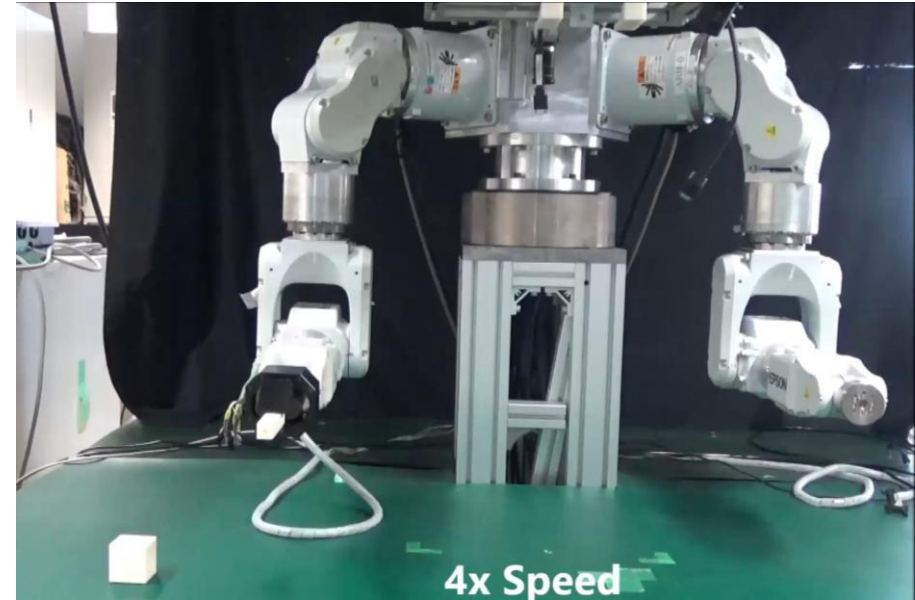
Scholarship: Japanese Government MEXT Scholarship.

Thesis Topic: Assembly Planning Using Rapidly Exploring Random Trees (RRT) Method.

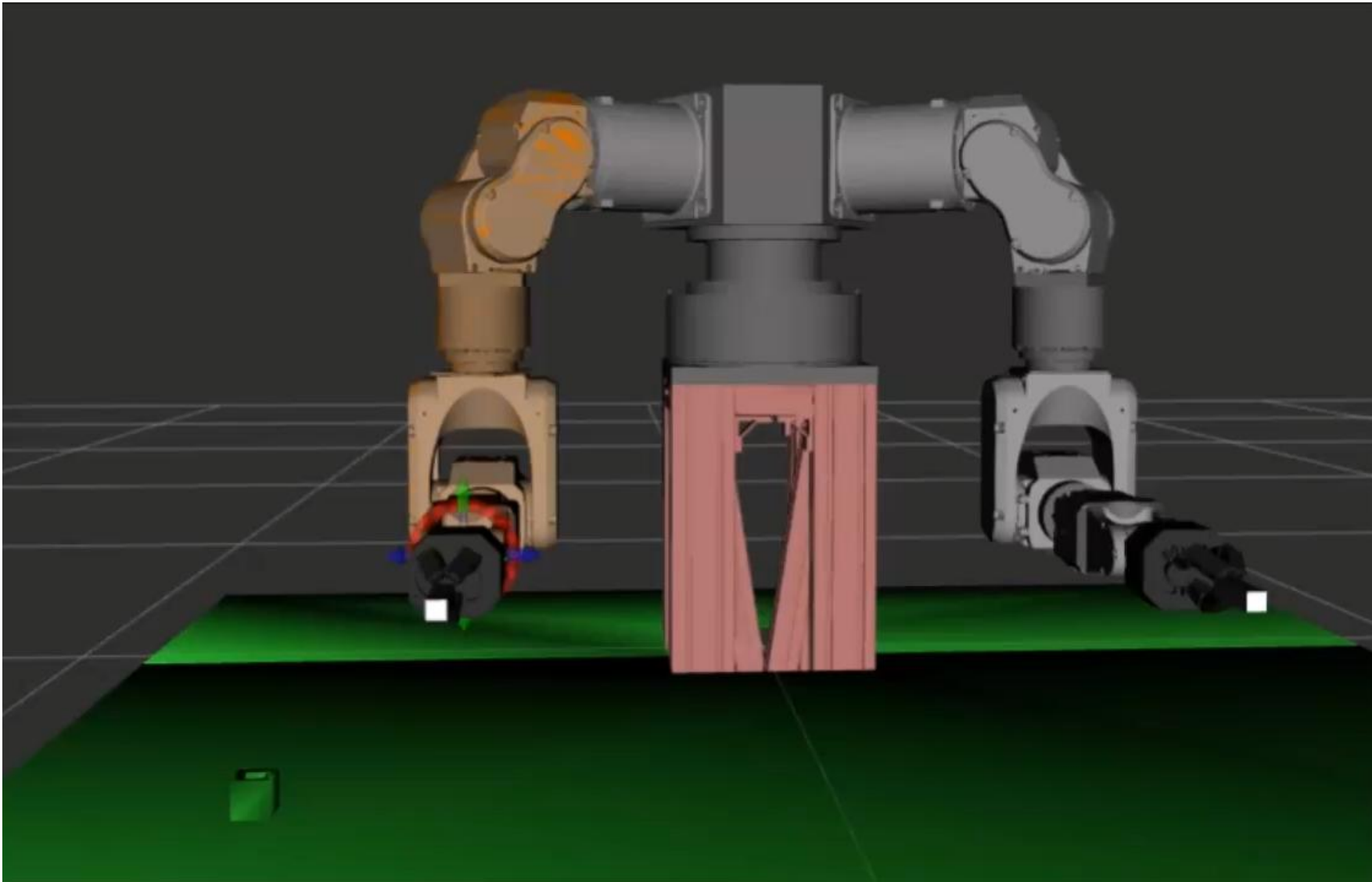
Funding: DENSO, Japan.

Supervisors: Prof. Kazuhiro Kosuge, Dr. Shogo Arai.

Key Areas: Assembly Planning, Motion Planning, Industrial Robots.



The robot performs peg-in-a-hole assembly problem



Institute: Tohoku University, Japan.

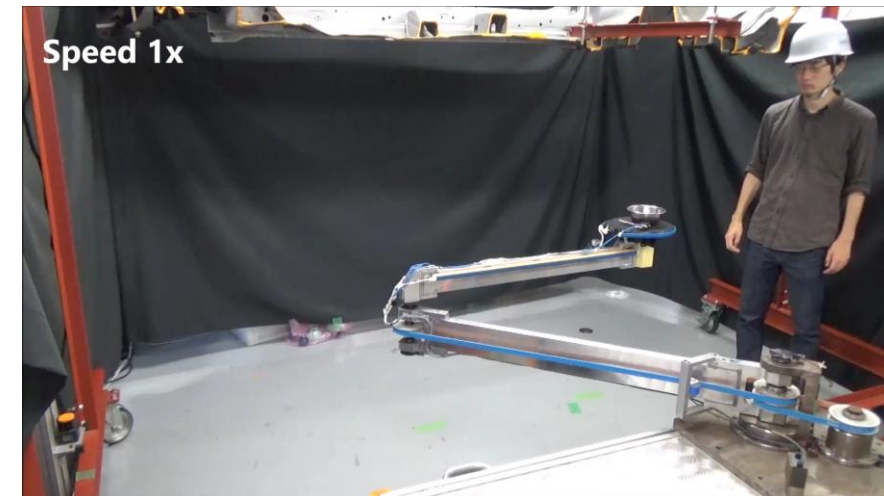
Scholarship: Japanese Government MEXT Scholarship.

Thesis Topic: Motion Planning for Collaborative Robots Using HRI-based Cost Function.

Funding: Toyota, Japan.

Supervisors: Prof. Kazuhiro Kosuge, Dr. Yasuhisa Hirata.

Key Areas: Collaborative Robots, Motion Planning, Human-Robot Interaction (HRI), Robot Control.



The robot provides assistance to a factory worker

PG Courses Taught:

1. Advanced Robotics with ROS (Spring 26)
2. Augmented and Virtual Reality (Fall 25)
3. Reinforcement Learning (Fall 24)
4. Robot Mechanics and Control (Spring 24)
5. Simultaneous Localization and Mapping (SLAM) – (Fall 21, Fall 22 & Fall 23).
6. Robot Design (Spring 22, Spring 25).
7. Research Methodology (Spring 22).
8. Engineering Mechanics (Fall 23).

UG Courses Taught:

1. Control Engineering (Spring 25)
2. Applications of ICT (Fall 24, Fall 25)
3. Fundamentals of Programming-I (Fall 23, Spring 26).
4. Robotics and Automation (Fall 21).
5. Electrical Engineering (Fall 21, Fall 22).
6. Electronics Engineering (Spring 22).

MS Thesis Supervision:

Sr.	Student Name	Batch	Thesis Topic	Duration
20.	Muhammad Riaz	RIME 2023	Explainable Multimodal Deep Learning for Early Detection of Autism Among Children	Sep 2025 - Present
19.	Fatima Nadeem	RIME 2023	Comparative Analysis of Dynamics Algorithms For Robotic Platforms	June 2025 - Present
18.	Muhammad Haris Zaman	RIME 2023	Corrosion Inspection Based on Point Cloud Data	May 2025 - Present
17.	Safi ur Rehman	RIME 2023	Computer Vision based Weeding for Agricultural Robots	May 2025 - Present
16.	Jaysh Muhammad	RIME 2023	Robotic Training using Generative AI	May 2025 - Present
15.	Mesum Ali Irfani	RIME 2023	Obstacle Avoidance for Marine Robots	May 2025 - Present
14.	Aina Zulfiqar	RIME 2023	Estimation of Psychological Parameters during Human Robot Collaboration	May 2025 - Present
13.	Umna Liaquat	BME 2023	Social Media Analytics for Mental Health Assessment	Mar 2025 - Present
12.	Tashfeen Ali Khan	RIME 2023	Visual SLAM in Dynamic Environments	Mar 2025 - Present
11.	Shakeel Ahmed	RIME 2021	Control of Flywheel Inverted Pendulum Using Reinforcement Learning	Sep 2023 - Jul 2025

MS Thesis Supervision:

Sr.	Student Name	Batch	Thesis Topic	Duration
10.	Ayesha Razzaq	BME 2021	Drug Discovery Using Generative AI	Aug 2024 - Aug 2025
9.	Muhammad Awais Saleem	RIME 2022	Physics Inspired Machine learning Simulator for Heating and Insulation Effects on houses	Jan 2024 - Sep 2025
8.	Muhammad Uzair Memon	RIME 2021	Assessment of Cognitive load in Augmented Reality Based Mobile Robot Navigation	Dec 2023 - Aug 2025
7.	Khubaib Haider Khan	RIME 2021	Addressing the Problem of Sloshing in a Liquid Carrying Mobile Robot Through AI	Jun 2023 - Aug 2025
6.	Soban Ahmed	RIME 2020	Development of Safety/Ethical Framework for Metaverse	Jan 2023 - Oct 2024
5.	Zahra Khan	RIME 2021	Robot Motion Planning Using Deep Reinforcement Learning	Sep 2022 - Aug 2025
4.	Saad Jamil	RIME 2021	Impact Dynamics for Humanoid Robot	Jul 2022 - Nov 2024
3.	Muhammad Salman Sadiq	RIME 2020	Fast Marching Trees (FMT*) for Dynamic Motion Planning	May 2022 - Oct 2024
2.	Izma Naveed	RIME 2020	Augmented Reality Based SLAM	Feb 2022 - Aug 2024
1.	Zainullah Khan	RIME 2020	Gait Generation for Quadrupedal Robots	Jan 2022 - Oct 2024

BE FYP Supervision:

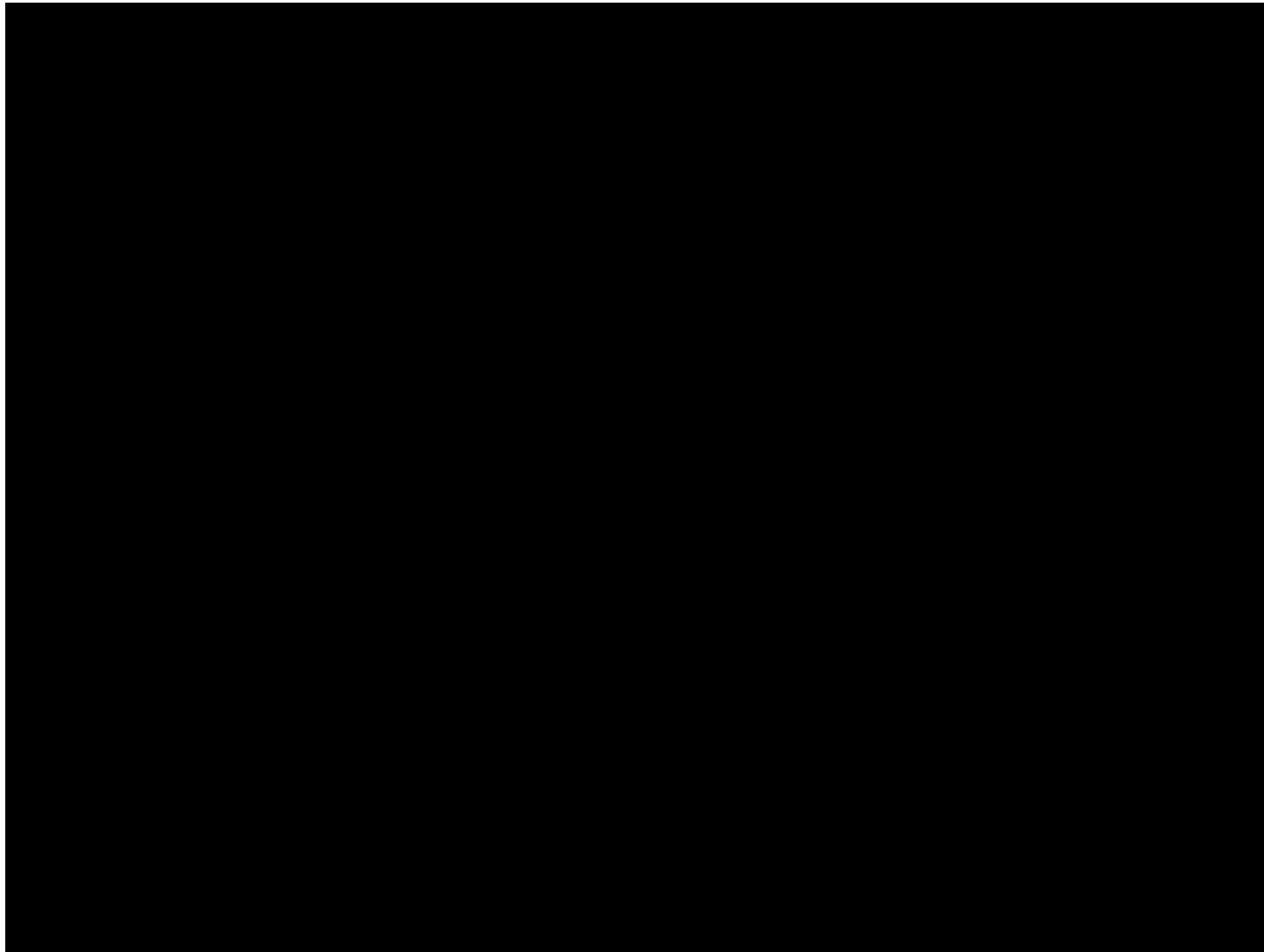
Sr.	Student Name	Batch	Thesis Topic	Duration	Achievement
9.	Muhammad Azhaf Zeeshan, Zainab Hameed, Mowahhid Bin Safdar, Muhammad Zain Ul Abideen	SMME ME 2022	Wall Climbing Robot for Surface Defect Detection	Apr 2025 - Present	
8.	Muhammad Huzaifa, Wajiha Ali	SEECs EE 2021	AI-based Frailty Assessment System for the Older Adults	Sep 2024 – May 2025	Journal Paper (IF 5.0), Documentary on NHK World
7.	Muhammad Talha Talib, Moosa Jillani, Muhammad Nuzran, Ahmed Bilal	SMME ME 2021	Design and Development of a Cricket Bowling Machine	June 2024 – May 2025	
6.	Muhammad Haris, Hamza Hussain, Osama Jarrar, Muhammad Saad bin Asif	SMME ME 2020	4-DOF Robotic Arm for Industrial Automation	Sep 2023 – May 2024	1st Prize at SMME Open House 2024
5.	Hinata Kashif, Jawad Akbar	SMME ME 2020	Design and Development of a Quadruped Robot	Sep 2023 – May 2024	Won IGNITE NGIRI Funding

BE FYP Supervision:

Sr.	Student Name	Batch	Thesis Topic	Duration	Achievement
4.	Somaan Shakeel, Ahmed Ali Siddiqui Muhammad Ahsan Saleem, Muhammad Hamza	SMME ME 2020	5-Axis Parallel Kinematic Robot	Sep 2023 – May 2024	
3.	Muhammad Nouman Aqil, Ali Wasi Awan, Fizza Arif, Abdul Moiz	SMME ME 2019	Development of Mobile Robot for Warehouse Automation	Sep 2022 – June 2023	Won IGNITE NGIRI Funding
2.	Muhammad Ibrahim Khan, Ahmad Hasan, Muhammad Usama Hannan, Muhammad Shaaf Nadeem	SMME ME 2019	Autonomous Weeding Mobile Robot	Sep 2022 – Jun 2023	3rd Prize at SMME Open House 2023
1.	Muhammad Abdullah Younas, Muhammad Ammar, Muhammad Moeed Ahmed, Khezar Qayyum	SMME ME 2018	Autonomous Mobile Robot with Chain-Driven Live Roller Top Module	Sep 2021 – Jun 2022	Conference Paper, Won IGNITE NGIRI Funding

- Robocup is the football worldcup of robots.
- The aim of Robocup is to develop a robot team that will be able to defeat the FIFA Worldcup champions by the year 2050.
- We are the first team from South Asia to qualify for the Robocup Standard Platform league; the most prestigious competition of Robocup.

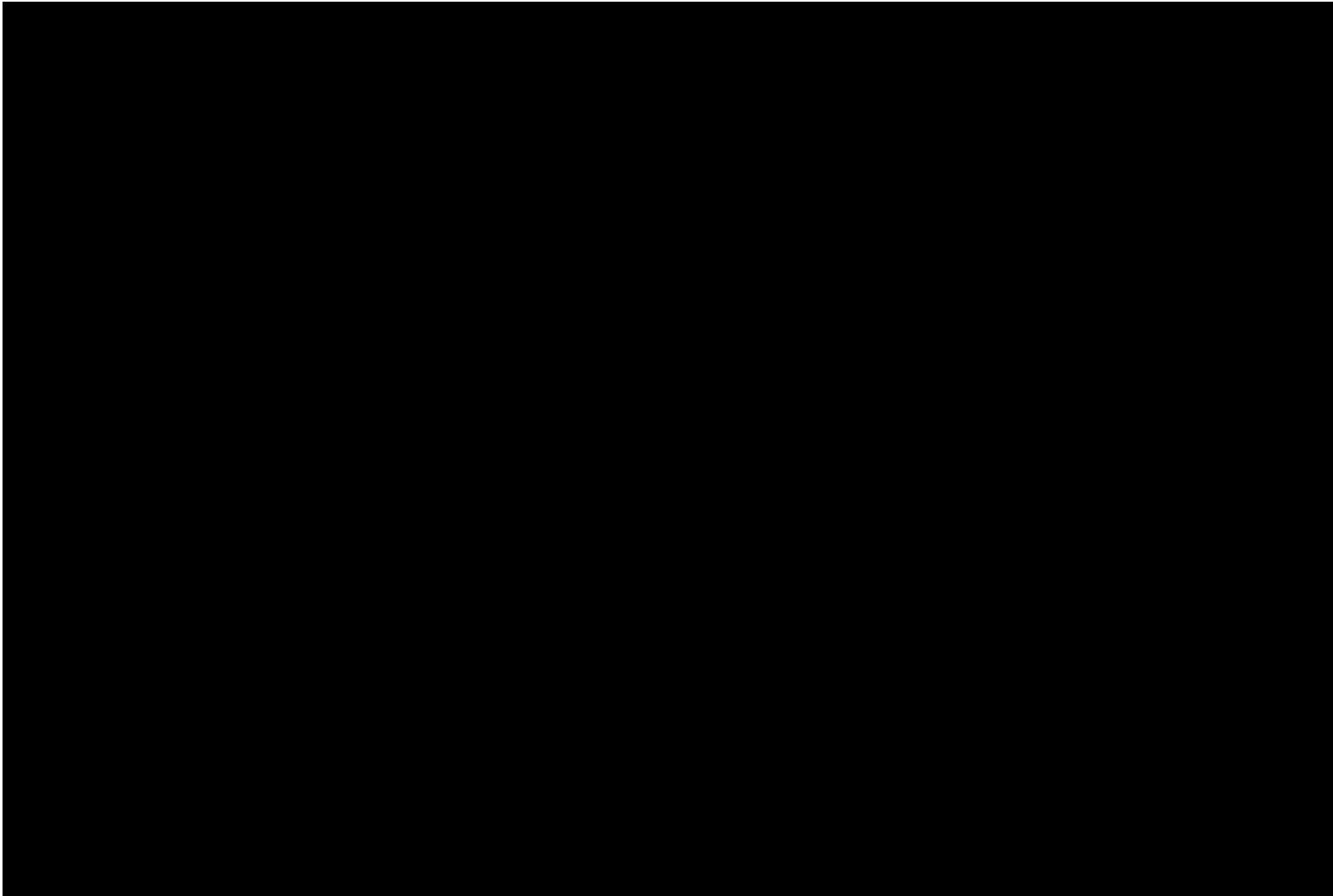




Robocup Standard Platform League

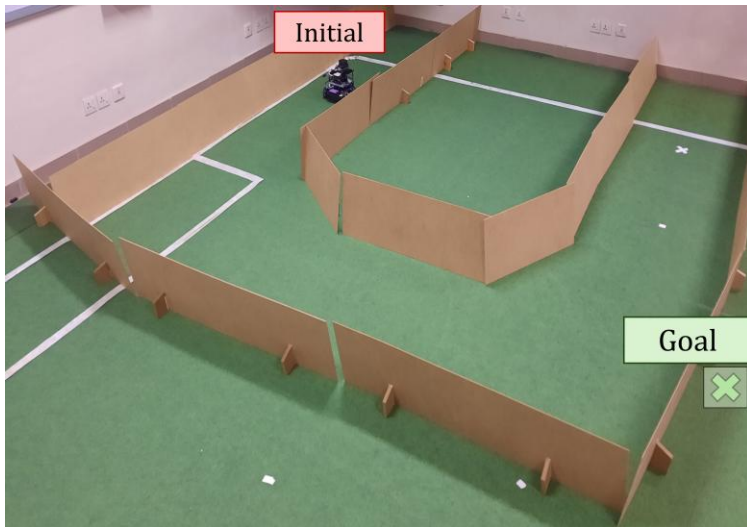
- Continuation of my FYP at SMME.
- Conversion of an electronic wheelchair to a robotic wheelchair
- Collaboration with Sakura wheelchair; a consortium of 7 wheelchair manufacturing companies in Japan.
- Three input modes: joystick, eyeball tracking, EEG.





Sakura Wheelchair Project

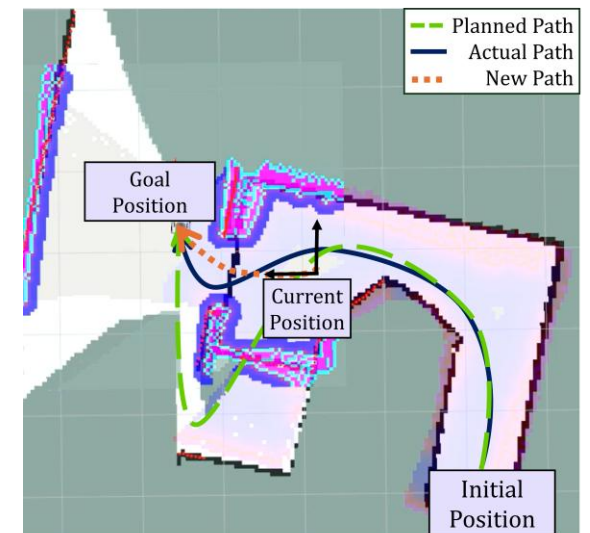
- Integration of AR with Hector SLAM to generate a map of the environment in real-time.
- The resulting map information, LIDAR data, and odometry are used in a motion controller that generated trajectories that were found to be much safer and smoother as compared to the existing state-of-the-art.
- In particular, there was a significant decrease in jerk and stress for all three experimental scenarios under consideration.



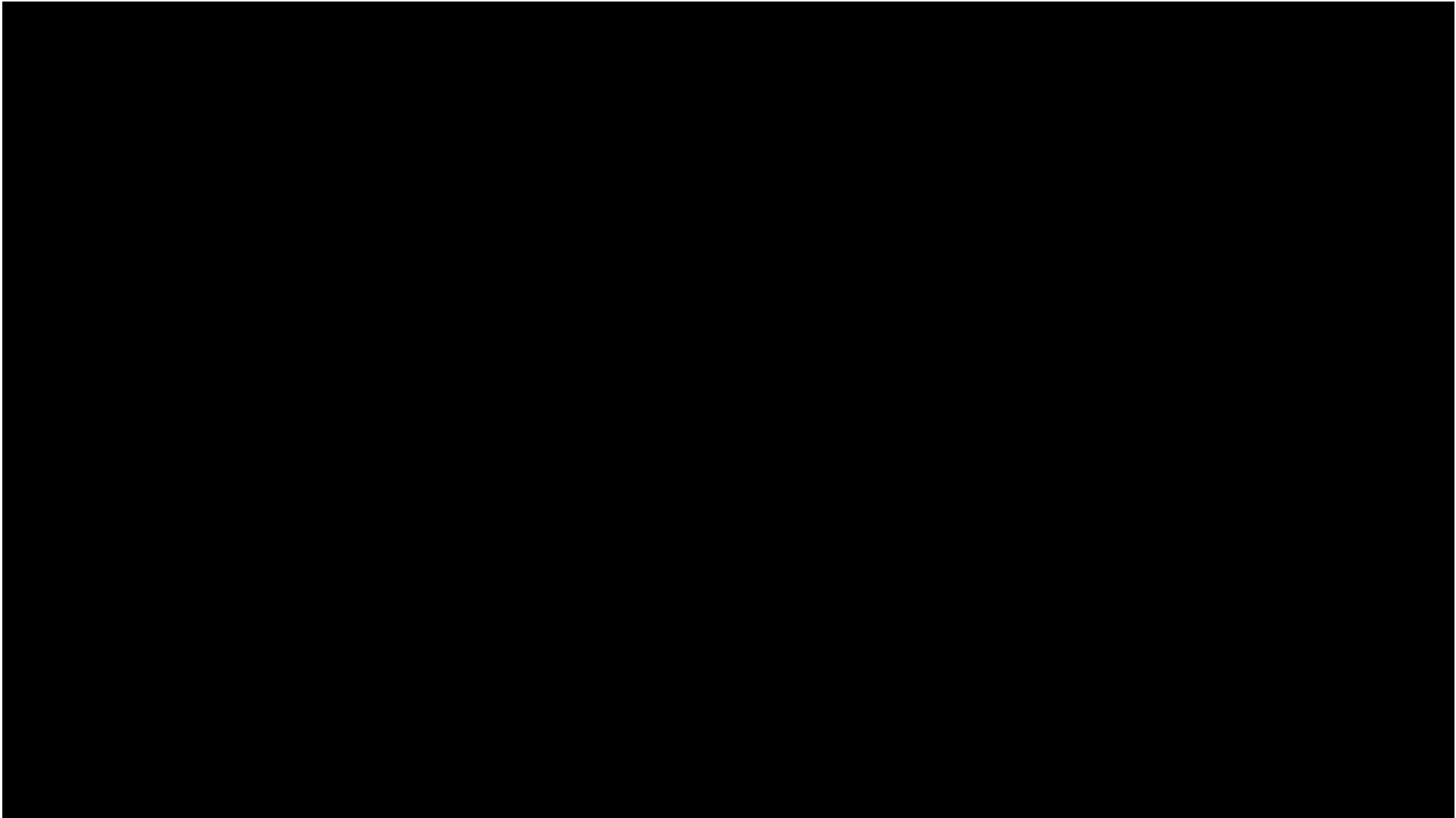
Real-world set up



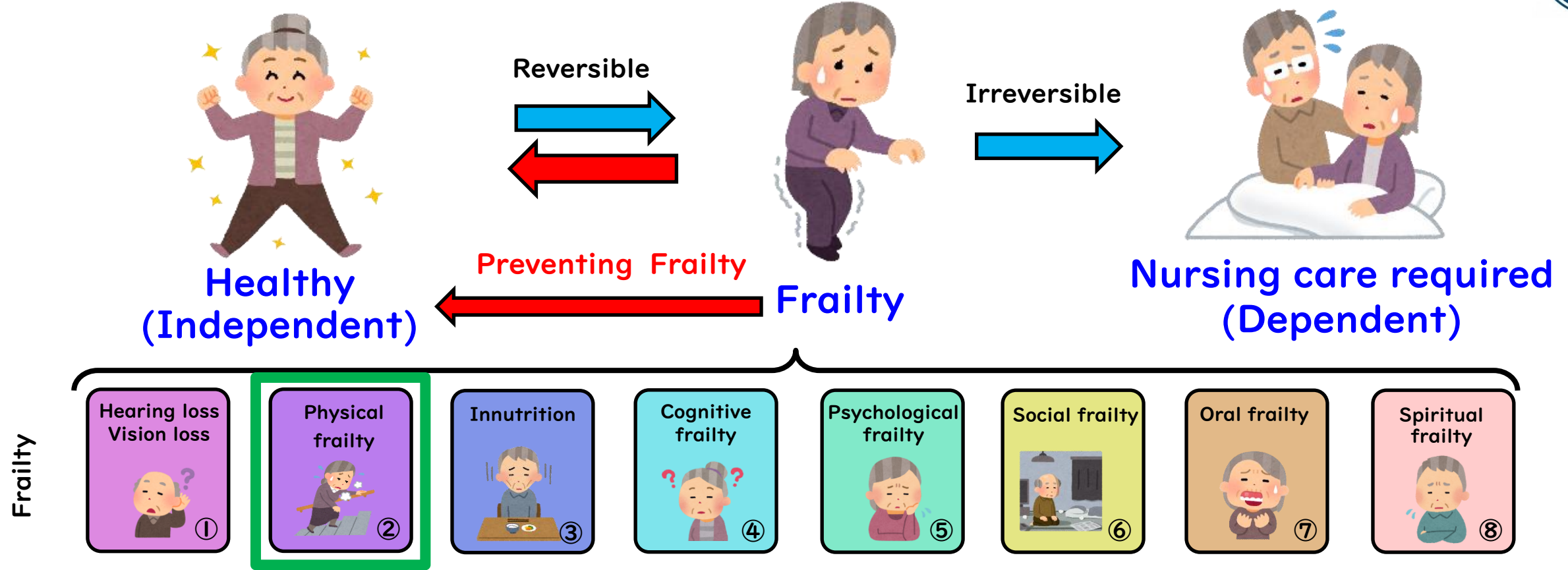
Simulation Environment



Navigation View



Augmented Reality based SLAM



- We have categorized frailty in eight domains to be holistically assessed.
- It may take more than four hours to assess whole domains using existing instruments.
- The aim is to assess whole domains in less than one hour using IT & AI technologies.

Part I:

AI-based visual measurement software for assessment of physical frailty

Long-seated forward bend (flexibility)



Functional reach (dynamic balance ability)



Standing on one leg with eye open (Equilibrium ability)



Grip strength (muscle Strength)

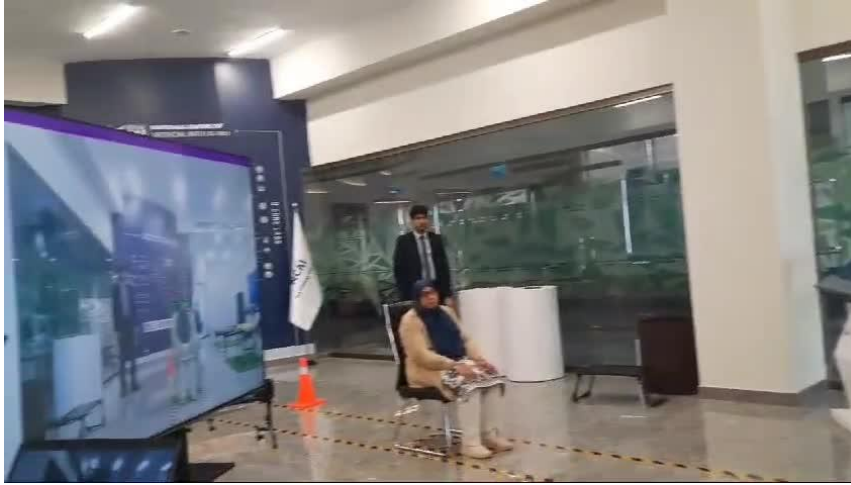


Walking speed (ability to move)



Timed Up & Go (ability to move)









- **Collaborative Robots**
- **Motion Planning**
- **Embodied AI**
- **Human-Robot Interaction**
- **Simultaneous Localization and Mapping (SLAM)**
- **Reinforcement Learning**
- **Virtual and Augmented Reality**
- **Generative AI**

- Franka Research 3 Collaborative Robot
- DJI Mavic 3 Drone
- EFT G630 Agricultural Drone
- Turtle Bot 4
- Core i7 11th Generation PCs with RTX 4070



- My Introduction
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- **Lectures (3 credit hours):**
 - 3x Assignments = 9%
 - 6x Quizzes (Announced + Surprise) = 9%
 - 2x Projects = 12%
 - Class Participation = 5%
 - Mid-Semester Exam = 25%
 - End Semester Exam = 40%

Class Participation:

- This is a measure of how involved you are in the class.
- I will ask a lot of questions during the lecture. If you answer these question correctly, you will get class participation points.
- If you demonstrate a negative attitude towards learning or cause a disturbance in the class, I will deduct class participation points.



- I will mark the attendance 15-20 minutes after the start of the lecture. Anyone who arrives after the attendance will be considered absent from the whole lecture.
- Don't talk amongst yourselves in the class. If you have any question, raise your hand and ask me.
- When I ask any question in the class, don't speak out of turn. Raise your hand and wait for me to get to you.
- If I catch you using your mobile phone during the lecture, then you will be fined **Rs 500**. This fine will be collected by the CR and will be used to arrange a party during the last lecture of the course.

- The main topics that we are going to cover in this course are:
 1. Introduction to Robotics and ROS
 2. ROS Tools and Simulation
 3. Robot Perception
 4. Robot Motion Planning and Control
 5. Advanced ROS Topics

- My Introduction
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Questions:

1. Where did you grow up? What is your city/region famous for?
2. Tell us your educational background (any achievements?).
3. Any experience in Robotics and ROS? (courses, projects, research?)
4. Are you a part time student or full time?
5. Why did you choose NUST and R&AI? Are you satisfied with your experience here so far?
6. What do you want to achieve in life?

Thank you for your attention.

Any Questions?

